

Name: _____

This practice exam covers Chapters 3, 4 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

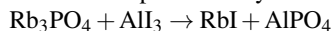
1. A particular planet has a mass of 2.72×10^{27} g and an oil tanker has a mass of 6.99×10^6 kg. How many moles of oil tankers have a mass equal to the mass of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

2. A particular planet has a mass of 3.76×10^{26} g and a cooking pot has a mass of 2.2×10^{-1} kg. What is the mass of 1 mole of cooking pots? Ensure your answer has the correct number of significant figures.

ANSWER: _____

3. A way to produce solid aluminum phosphate is by combining aqueous rubidium phosphate with aqueous aluminum iodide. The chemical reaction is represented by the following unbalanced chemical equation:



Patricia wants to produce 14.80 mg of aluminum phosphate. She pours 15.8 mL of a 0.03287-M solution of aqueous rubidium phosphate into a beaker. What volume of a 0.03110-M solution of aqueous aluminum iodide is needed to react with the rubidium phosphate solution to produce 14.80 mg of aluminum phosphate? If you find that 15.8 mL of the rubidium phosphate solution is insufficient, report the required volumes of both reactants required to produce the 14.80 mg of aluminum phosphate.

ANSWER: _____

4. Measurements show that an unknown compound has the following composition:

Element	Mass %
Cs	51.15 %
I	48.85 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

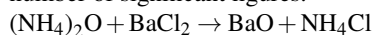
5. Calculate the mass percent of oxygen in silver chromate (Ag_2CrO_4). Round your answer to the nearest percentage.

ANSWER: _____

6. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

7. Given this unbalanced chemical equation, report the mass of $(\text{NH}_4)_2\text{O}$ needed to produce 6.446 g of BaO . Ensure your answer has the correct number of significant figures.



ANSWER: _____

8. Matthew is instructed to dissolve 2.69 g of barium iodide in 140.4 mL of a 94.7-mM aqueous solution of ammonium fluoride. Calculate the molarity of the fluoride anion in the solution after the reaction has occurred. Assume the volume of the solution does not change when fluoride is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

9. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

10. Thomas weighs out 1.97 g of an unknown diprotic acid into a 300.-mL volumetric flask and dilutes the solution to the mark with distilled water. He then titrates this solution with 0.351-M rubidium hydroxide solution. When the titration reaches the equivalence point, Thomas finds that he has added 329.41 mL of the rubidium hydroxide solution. Calculate the molar mass of the unknown diprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

11. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

12. Scott conducted some measurements and determined that 0.628 moles of francium participated in a chemical reaction. Calculate the mass of francium that participated in the chemical reaction.

ANSWER: _____

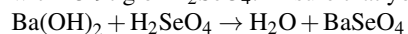
13. An oil tanker has a mass of 2.2×10^7 kg and an atom of iron has a mass of 9.62×10^{-23} g. How many moles of oil tankers have a mass equal to the mass of an atom? Ensure your answer has the correct number of significant figures.

ANSWER: _____

14. As part of her routine quality-control checks, Chloe, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of hydrofluoric acid solution. She takes a 40.0-mL sample of hydrofluoric acid solution and titrates it with a sodium hydroxide solution, which she made by dissolving 3.648 g of solid sodium hydroxide in 1.377 L of distilled water. If Chloe finds that it takes 585.3 mL of the sodium hydroxide solution for the titration to reach the equivalence point, what molarity of the hydrofluoric acid solution does Chloe calculate?

ANSWER: _____

15. Given this unbalanced chemical equation, report the maximum mass of BaSeO_4 that could be produced by the mixing of 11.49 g of Ba(OH)_2 with 15.97 g of H_2SeO_4 . Ensure that your answer has the correct number of significant figures.

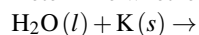


ANSWER: _____

16. Ammonium sulfide reacts with chromium (III) chloride to form chromium (III) sulfide and ammonium chloride. Suppose 22.40 g of ammonium sulfide is mixed with 22.65 g of chromium (III) chloride. Calculate and report the theoretical yield of ammonium chloride formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

17. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".



ANSWER: _____

18. Nicholas prepares a solution of chromium (III) iodide (CrI_3) by measuring out 5.70 g of chromium (III) iodide into a 50.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

19. Paul measures the amount of lithium oxide produced during a chemical reaction. He finds that 0.5636 mol is produced. Calculate the mass of lithium oxide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

20. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 152 kg of compound A and 248 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 88.34%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

21. Paul weighs out 1.017 g of an unknown diprotic acid into a 250.-mL volumetric flask and dilutes the solution to the mark with distilled water. He then titrates this solution with 0.3927-M aluminum hydroxide solution. When the titration reaches the equivalence point, Paul finds that he has added 27.836 mL of the aluminum hydroxide solution. Calculate the molar mass of the unknown diprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

22. Joseph measures the amount of ammonium chloride produced during a chemical reaction. He finds that 509.41 g is produced. Calculate the number of moles of ammonium chloride produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

23. John conducted some measurements and determined that 439. g of oxygen gas participated in a chemical reaction. Calculate the number of moles of oxygen gas that participated in the chemical reaction.

ANSWER: _____

24. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

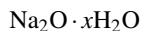
25. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

26. Copper (II) phosphate reacts with ammonium bromide to form copper (II) bromide and ammonium phosphate. Calculate the moles of copper (II) phosphate needed to produce 1.374 moles of copper (II) bromide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

27. Sodium oxide forms several hydrates with the general form



where x is an integer. If the hydrate is heated, the water can be driven off, leaving pure Na_2O behind. Suppose a sample of a certain hydrate of sodium oxide is heated until all the water is removed, and a chemist finds that the mass of the sample decreases by 54%. Give the value of x to indicate which hydrate the sample was.

ANSWER: _____

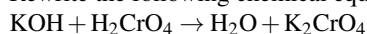
28. Hydronium sulfide reacts with barium chloride to form barium sulfide and hydronium chloride. Suppose 3.70 g of hydronium sulfide is mixed with 17.7 g of barium chloride. Calculate the maximum mass of barium sulfide that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

29. What major chemical species are present in an aqueous solution of 0.0155-M hydronium oxide?

ANSWER: _____

30. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.



ANSWER: _____

31. Measurements show that an unknown compound has the following composition:

Element	Mass %
Hydrogen	2.19 %
Oxygen	58.0 %
Manganese	39.8 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

32. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($Ca(NO_3)_2$)	11.0%
urea ($(NH_2)_2CO$)	38.0%
potassium sulfate (K_2SO_4)	29.0%
ammonium phosphate ($(NH_4)_3PO_4$)	13.0%
inert ingredients	9.0%

What should the second (potassium) number on the bag's label be? In other words, calculate the percent by mass of potassium in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

33. Dylan prepares a solution of aluminum chloride ($AlCl_3$) by measuring out 26.33 g of aluminum chloride into a 100.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

34. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($Ca(NO_3)_2$)	26.0%
urea ($(NH_2)_2CO$)	19.0%
potassium sulfate (K_2SO_4)	22.0%
ammonium phosphate ($(NH_4)_3PO_4$)	26.0%
inert ingredients	7.0%

What should the second (potassium) number on the bag's label be? In other words, calculate the percent by mass of potassium in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

35. Which element is 48% (by mass) of aluminum dichromate ($Al_2(Cr_2O_7)_3$)? Show your work.

ANSWER: _____

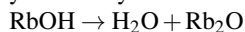
36. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 565 kg of compound A and 228 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 85.66%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

37. A strip of solid strontium metal is put into a beaker of 0.32-M $\text{Cu}(\text{BrO})_2$ solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

38. As RbOH is converted into H_2O in the following reaction, determine whether hydrogen is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the hydrogen found in RbOH and H_2O .



ANSWER: _____

39. Joseph must dilute 128.6 mL of a 2.2-mM solution of aqueous aluminum bisulfate ($\text{Al}(\text{HSO}_4)_3$) until the concentration falls to 1.0 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

40. As part of his routine quality-control checks, Adam, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of nitric acid solution. He takes a 50.0-mL sample of nitric acid solution and titrates it with a sodium hydroxide solution, which he made by dissolving 1.26 g of solid sodium hydroxide in 1.331 L of distilled water. If Adam finds that it takes 352 mL of the sodium hydroxide solution for the titration to reach the equivalence point, what molarity of the nitric acid solution does Adam calculate?

ANSWER: _____

41. Jason prepares a solution of potassium fluoride (KF) by measuring out 36.1 mmol of potassium fluoride into a 100.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

42. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

43. Of the following labels, identify all that can be applied to the chemical compound B_2Te_3 : ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

44. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

45. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

46. Given this unbalanced chemical equation, report the maximum mass of H_2O that could be produced by the mixing of 59.21 g of KOH with 11.17 g of H_2S . Ensure that your answer has the correct number of significant figures.
- $$\text{KOH} + \text{H}_2\text{S} \rightarrow \text{H}_2\text{O} + \text{K}_2\text{S}$$

ANSWER: _____

47. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

48. Allison weighs out 1.270 g of hypochlorous acid (HClO) into a 50.-mL volumetric flask and dilutes the solution to the mark with distilled water. She plans to titrate the acid with 0.385-M strontium hydroxide solution. Calculate the volume of strontium hydroxide solution that Allison must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

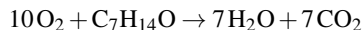
49. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

50. Sophia has a 1.366-M solution of strontium bisulfate ($\text{Sr}(\text{HSO}_4)_2$). If she needs to add 0.315 g of strontium bisulfate to a reaction flask, what volume of the solution should she pour? Be sure your answer has the correct number of significant figures.

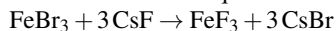
ANSWER: _____

51. What is the total number of carbon atoms shown on the reactant side of this chemical equation?



ANSWER: _____

52. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:



ANSWER: _____

53. Allison conducted some measurements and determined that 0.202 moles of bromine liquid participated in a chemical reaction. Calculate the mass of bromine liquid that participated in the chemical reaction.

ANSWER: _____

54. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

55. Given this unbalanced chemical equation, calculate and report the percent yield of FeF_3 if 12.6 g of FeF_3 is produced from the reaction of 25.0 g of Fe_2S_3 and 18.30 g of H_3OF . Ensure that your answer has the correct number of significant figures.
- $$\text{Fe}_2\text{S}_3 + \text{H}_3\text{OF} \rightarrow \text{FeF}_3 + (\text{H}_3\text{O})_2\text{S}$$

ANSWER: _____

56. Shiv is performing a combustion analysis on a compound with a molecular molar mass of 26.04 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 3.62 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	12.24 g
Water	2.5 g

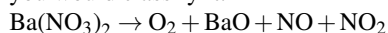
Use this information to find the molecular formula of the compound.

ANSWER: _____

57. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.
- $$\text{Br}_2 + \text{Na} \rightarrow \text{NaBr}$$

ANSWER: _____

58. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

59. Chloe puts a small sample of Cs_2S in water and lets it dissolve. After equilibrium is reached, she determines that several chemical species remain in the solution. These include Cs^+ , S^{2-} , and H_2O . Which of the following labels apply to the compound Cs_2S ? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

60. Ethan adds 240.6 mL of a 1.76-M solution of rubidium sulfide (Rb_2S) to a reaction flask. Calculate the amount of rubidium sulfide that he has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

61. Aluminum bromide reacts with ammonium oxide to form aluminum oxide and ammonium bromide. Suppose 124 g of aluminum bromide is mixed with 26.5 g of ammonium oxide. Calculate and report the theoretical yield of aluminum oxide formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

62. Eric conducted some measurements and determined that 0.150 moles of moscovium participated in a chemical reaction. Calculate the mass of moscovium that participated in the chemical reaction.

ANSWER: _____

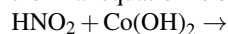
63. Dylan prepares a solution of lithium phosphate (Li_3PO_4) by measuring out 16.22 mmol of lithium phosphate into a 500.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

64. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

65. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

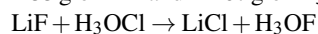
66. Hydroiodic acid reacts with caesium hydroxide to form water and caesium iodide. Suppose 118.7 g of hydroiodic acid is mixed with 74.56 g of caesium hydroxide. Calculate and report the theoretical yield of water formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

67. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

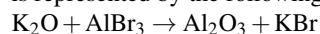
ANSWER: _____

68. Given this unbalanced chemical equation, calculate and report the percent yield of LiCl if 6.99 g of LiCl is produced from the reaction of 4.55 g of LiF and 11.67 g of H_3OCl . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

69. A way to produce solid aluminum oxide is by combining aqueous potassium oxide with aqueous aluminum bromide. The chemical reaction is represented by the following unbalanced chemical equation:



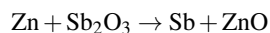
Knowing this, Joseph pours 15.82 mL of a 0.0749-M solution of aqueous potassium oxide and then adds 11.01 mL of a 0.15-M solution of aqueous aluminum bromide. He then stirs the mixture and notices aluminum oxide precipitating out of the solution. What is the mass of aluminum oxide that he should expect the reaction to yield?

ANSWER: _____

70. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

71. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.

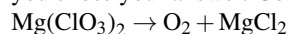


ANSWER: _____

72. Suppose Victoria heats a sample of a certain chromium (III) oxide hydrate, $\text{Cr}_2\text{O}_3 \cdot 9\text{H}_2\text{O}$ until all the water is removed and pure Cr_2O_3 is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

73. As $\text{Mg}(\text{ClO}_3)_2$ is converted into O_2 in the following reaction, determine whether oxygen is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the oxygen found in $\text{Mg}(\text{ClO}_3)_2$ and O_2 .



ANSWER: _____

74. Linda conducted some measurements and determined that 0.545 moles of fluorine gas participated in a chemical reaction. Calculate the mass of fluorine gas that participated in the chemical reaction.

ANSWER: _____

75. Brian must prepare 764 mL of a 0.754-mM working solution of aqueous caesium sulfide (Cs_2S). He will do this by pouring some 0.033589-M aqueous caesium sulfide stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-caesium sulfide stock solution must Brian pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

76. Michael measures the amount of antimony trichloride produced during a chemical reaction. He finds that 0.9865 mol is produced. Calculate the mass of antimony trichloride produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

77. $\text{Mg}_3(\text{PO}_4)_2$ is the formula for magnesium phosphate. Benjamin, a chemist, measures a sample of magnesium phosphate and determines that it contains 0.2596 moles of phosphorus. How many moles of oxygen are in the sample?

ANSWER: _____

78. Calculate the number of sulfur atoms in a 9.001×10^{-3} -g sample of selenium monosulfide (SeS).

ANSWER: _____

79. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

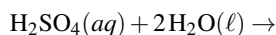
80. A brick has a mass of 6.55×10^0 kg and a coin has a mass of 4.3×10^0 g. What is the mass of 1 mole of bricks? Ensure your answer has the correct number of significant figures.

ANSWER: _____

81. Anna adds 223.25 mL of a 1.392-M solution of chromium (III) iodide (CrI_3) to a reaction flask. Calculate the amount of chromium (III) iodide that she has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

82. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.



ANSWER: _____

83. Sam weighs out 2.053 g of an unknown monoprotic acid into a 25-mL volumetric flask and dilutes the solution to the mark with distilled water. He then titrates this solution with 0.3085-M caesium hydroxide solution. When the titration reaches the equivalence point, Sam finds that he has added 182.53 mL of the caesium hydroxide solution. Calculate the molar mass of the unknown monoprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

84. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_3B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 0.7110 moles of A and 0.474 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

85. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 83.2 g.

Compound	Initial Mass	Final Mass
A	40.5 g	11.7 g
B	79.3 g	0 g

ANSWER: _____

86. David conducted some measurements and determined that 0.132 moles of mercury participated in a chemical reaction. Calculate the mass of mercury that participated in the chemical reaction.

ANSWER: _____

87. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_2B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 3.48 moles of A and 6.962 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

88. Zinc hydroxide reacts with nitric acid to form water and zinc nitrate. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

89. Measurements show that an unknown compound has the following composition:

Element	Mass %
Nitrogen	26.2 %
Hydrogen	7.54 %
Chlorine	66.3 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

90. Shiv prepares a solution of hydronium phosphate ($(\text{H}_3\text{O})_3\text{PO}_4$) by measuring out 29.35 mmol of hydronium phosphate into a 50.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

91. Calculate the mass percent of chromium in copper (II) dichromate (CuCr_2O_7). Round your answer to the nearest percentage.

ANSWER: _____

92. Lithium fluoride reacts with hydronium sulfide to form lithium sulfide and hydronium fluoride. Calculate the mass of hydronium sulfide needed to produce 12.4 g of lithium sulfide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

93. Jennifer weighs out 2.16 g of arsenous acid (H_3AsO_3) into a 25-mL volumetric flask and dilutes the solution to the mark with distilled water. She plans to titrate the acid with 0.323-M iron (III) hydroxide solution. Calculate the volume of iron (III) hydroxide solution that Jennifer must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

94. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

95. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

96. $Ba(BrO)_2$ is the formula for barium hypobromite. Nick, a chemist, measures a sample of barium hypobromite and determines that it contains 2.31 moles of barium. How many moles of oxygen are in the sample?

ANSWER: _____

97. Given this unbalanced chemical equation, report the amount of AlI_3 needed to produce 0.167 moles of $AlBr_3$. Report your answer in moles. Ensure your answer has the correct number of significant figures.
- $$AlI_3 + CsBr \rightarrow CsI + AlBr_3$$

ANSWER: _____

98. Calculate the number of oxygen atoms in a 0.40-g sample of silicon dioxide (SiO_2).

ANSWER: _____

99. Chromium (III) iodide forms several hydrates with the general form $CrI_3 \cdot xH_2O$

where x is an integer. If the hydrate is heated, the water can be driven off, leaving pure CrI_3 behind. Suppose a sample of a certain hydrate of chromium (III) iodide is heated until all the water is removed, and a chemist finds that the mass of the sample decreases by 17%. Give the value of x to indicate which hydrate the sample was.

ANSWER: _____

100. Measurements show that an unknown compound has the following composition:

Element	Mass %
Sb	24.23 %
I	75.77 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

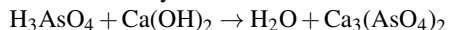
101. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 639 kg of compound A and 520. kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 94.34%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

102. An oil tanker has a mass of 1.7×10^7 kg and a particular moon has a mass of 7.09×10^{24} kg. How many moles of oil tankers have a mass equal to the mass of this particular moon? Ensure your answer has the correct number of significant figures.

ANSWER: _____

103. Given this unbalanced chemical equation, report the amount of H_3AsO_4 needed to produce 0.82 moles of $\text{Ca}_3(\text{AsO}_4)_2$. Report your answer in moles. Ensure your answer has the correct number of significant figures.



ANSWER: _____

104. An oil tanker has a mass of 2.36×10^8 kg and a brick has a mass of 1.30×10^0 kg. How many moles of oil tankers have a mass equal to the mass of a brick? Ensure your answer has the correct number of significant figures.

ANSWER: _____

105. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".
 $\text{H}_2\text{O}(l) + \text{Ba}(s) \rightarrow$

ANSWER: _____

106. Measurements show that an unknown compound has the following composition:

Element	Mass %
Carbon	13.65 %
Fluorine	86.35 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

107. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($\text{Ca}(\text{NO}_3)_2$)	22.0%
urea ($(\text{NH}_2)_2\text{CO}$)	25.0%
potassium sulfate (K_2SO_4)	24.0%
ammonium phosphate ($(\text{NH}_4)_3\text{PO}_4$)	23.0%
inert ingredients	6.0%

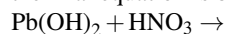
What should the third (phosphorus) number on the bag's label be? In other words, calculate the percent by mass of phosphorus in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

108. Robert needs to make 346.8 mL of aqueous lithium phosphate (Li_3PO_4) working solution. He plans to do this by adding distilled water to a 69.684-mL stock solution of 0.0183-M aqueous lithium phosphate. What will be the concentration of his working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

109. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

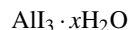
110. Joshua measures the amount of tetraantimony tricarbide produced during a chemical reaction. He finds that 1260 g is produced. Calculate the number of moles of tetraantimony tricarbide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

111. Victoria puts a small sample of HNO_3 in water and lets it dissolve. After equilibrium is reached, she determines that several chemical species remain in the solution. These include H_3O^+ , NO_3^- , and H_2O . Which of the following labels apply to the compound HNO_3 ? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

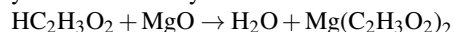
112. Aluminum iodide forms several hydrates with the general form



where x is an integer. If the hydrate is heated, the water can be driven off, leaving pure AlI_3 behind. Suppose a sample of a certain hydrate of aluminum iodide is heated until all the water is removed, and a chemist finds that the mass of the sample decreases by 26%. Give the value of x to indicate which hydrate the sample was.

ANSWER: _____

113. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

114. A strip of solid manganese metal is put into a beaker of 0.160-M NaClO_3 solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

115. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

116. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

117. Jason must dilute 104.0 mL of a 3.548-mM solution of aqueous rubidium phosphate (Rb_3PO_4) until the concentration falls to 1.782 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

118. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.
- $$\text{Be} + \text{CO}_2 \rightarrow \text{BeO} + \text{C}$$

ANSWER: _____

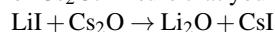
119. Calculate the mass percent of sodium in sodium chloride (NaCl). Round your answer to the nearest percentage.

ANSWER: _____

120. Brian is instructed to dissolve 4.931 g of copper (II) chloride in 215.0 mL of a 55.25-mM aqueous solution of ammonium phosphate. Calculate the molarity of the ammonium cation in the solution after the reaction has occurred. Assume the volume of the solution does not change when ammonium is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

121. Given this unbalanced chemical equation, calculate and report the theoretical yield of Li_2O formed by the mixing of 181.2 g of LiI with 269.1 g of Cs_2O . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

122. Boric acid reacts with bismuth hydroxide to form water and bismuth borate. Calculate the mass of bismuth hydroxide needed to produce 18.62 g of bismuth borate. Ensure your answer has the correct number of significant figures.

ANSWER: _____

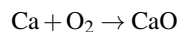
123. Sam adds 370.3 mL of a 0.23-M solution of chromium (III) nitrate ($\text{Cr}(\text{NO}_3)_3$) to a reaction flask. Calculate the amount of chromium (III) nitrate that he has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

124. Calculate the mass of a sample of silicon tetraiodide (SiI_4) that contains 1.16×10^{16} iodine atoms.

ANSWER: _____

125. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____

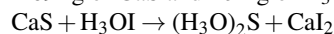
126. Calculate the mass of a sample of ammonium iodide (NH_4I) that contains 1.97×10^{13} nitrogen atoms.

ANSWER: _____

127. Jessica conducted some measurements and determined that 55.0 g of chlorine gas participated in a chemical reaction. Calculate the number of moles of chlorine gas that participated in the chemical reaction.

ANSWER: _____

128. Given this unbalanced chemical equation, calculate and report the percent yield of CaI_2 if 4.075 g of CaI_2 is produced from the reaction of 1.691 g of CaS and 10.1 g of H_3OI . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

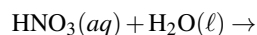
129. Adam conducted some measurements and determined that 990. g of iodine solid was produced in a chemical reaction. Calculate the number of moles of iodine solid that was produced in the chemical reaction.

ANSWER: _____

130. Aaron has a 0.158-M solution of hydronium oxide ($(\text{H}_3\text{O})_2\text{O}$). If he pours 54.3 mL of hydronium oxide into a reaction flask, what is the mass of hydronium oxide that he has added to the flask? Be sure your answer has the correct number of significant figures.

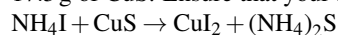
ANSWER: _____

131. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.



ANSWER: _____

132. Given this unbalanced chemical equation, calculate and report the theoretical yield of $(\text{NH}_4)_2\text{S}$ formed by the mixing of 85.9 g of NH_4I with 17.5 g of CuS . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

133. Adam puts a small sample of $\text{Ni}(\text{OH})_2$ in water and lets it dissolve. After equilibrium is reached, he determines that several chemical species remain in the solution. These include Ni^{2+} , OH^- , $\text{Ni}(\text{OH})_2$, and H_2O . Which of the following labels apply to the compound $\text{Ni}(\text{OH})_2$? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

134. The chemical formula for hydronium phosphate is $(\text{H}_3\text{O})_3\text{PO}_4$. Calculate the molar mass of hydronium phosphate. Round your answer to 5 significant figures.

ANSWER: _____

135. Isabella has a 0.8826-M solution of copper (II) cyanide ($\text{Cu}(\text{CN})_2$). If she pours 158 mL of copper (II) cyanide into a reaction flask, what is the mass of copper (II) cyanide that she has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

136. Chromium (III) phosphate reacts with iron (III) bromide to form chromium (III) bromide and iron (III) phosphate. Suppose 10.01 g of chromium (III) phosphate is mixed with 16.40 g of iron (III) bromide. Calculate the maximum mass of iron (III) phosphate that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

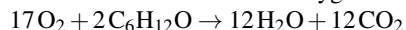
137. Jason conducted some measurements and determined that 1.09 g of terbium participated in a chemical reaction. Calculate the number of moles of terbium that participated in the chemical reaction.

ANSWER: _____

138. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

139. What is the total number of oxygen atoms shown on the product side of this chemical equation?



ANSWER: _____

140. The chemical formula for tellurium monoxide is TeO . Calculate the molar mass of tellurium monoxide. Round your answer to 5 significant figures.

ANSWER: _____

141. Measurements show that an unknown compound has the following composition:

Element	Mass %
Mg	27.7 %
P	23.6 %
O	48.7 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

142. Selenic acid reacts with copper (II) hydroxide to form water and copper (II) selenate. If 3.049 g of copper (II) selenate is produced from the reaction of 5.76 g of selenic acid and 6.63 g of copper (II) hydroxide, calculate the percent yield of copper (II) selenate. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

143. A strip of solid zinc metal is put into a beaker of 0.36-M $\text{Al}(\text{MnO}_4)_3$ solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

144. Aaron needs to make 227 mL of aqueous hydronium bromide (H_3OBr) working solution. He plans to do this by adding distilled water to a 68.244-mL stock solution of 0.0622-M aqueous hydronium bromide. What will be the concentration of his working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

145. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.
 $\text{CH}_3\text{OH} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$

ANSWER: _____

146. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".
 $\text{H}_2\text{O}(l) + \text{Rb}(s) \rightarrow$

ANSWER: _____

147. A virus has a mass of 8.73×10^{-11} mg and an atom of iron has a mass of 1.382×10^{-21} g. What is the mass of 1 mole of viruses? Ensure your answer has the correct number of significant figures.

ANSWER: _____

148. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

149. Samantha measures the amount of antimony monophosphide produced during a chemical reaction. She finds that 48.6 g is produced. Calculate the number of moles of antimony monophosphide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

150. Predict the products that form when sulfuric acid is mixed with lead (II) hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

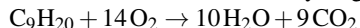
151. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_2B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 0.224 moles of A and 0.3362 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

152. Caesium hydroxide reacts with hydrosulfuric acid to form water and caesium sulfide. If 9.92 g of water is produced from the reaction of 139 g of caesium hydroxide and 11.8 g of hydrosulfuric acid, calculate the percent yield of water. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

153. What is the total number of hydrogen atoms shown on the reactant side of this chemical equation?



ANSWER: _____

154. Sophia must dilute 50.72 mL of a 3.008-mM solution of aqueous strontium chloride (SrCl_2) until the concentration falls to 0.6070 mM. She will do this by adding distilled water to the solution. How much distilled water must she add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

155. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

156. Which element is 20% (by mass) of trigermanium tetranitride (Ge_3N_4)? Show your work.

ANSWER: _____

157. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 86.7 g.

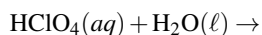
Compound	Initial Mass	Final Mass
A	36.8 g	7.4 g
B	79.0 g	0 g

ANSWER: _____

158. A virus has a mass of 9.62×10^{-11} mg and a particular moon has a mass of 9.5×10^{22} kg. What is the mass of 1 mole of this particular moon? Ensure your answer has the correct number of significant figures.

ANSWER: _____

159. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.

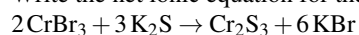


ANSWER: _____

160. The chemical formula for diphosphorus trisulfide is P_2S_3 . Calculate the molar mass of diphosphorus trisulfide. Round your answer to 4 significant figures.

ANSWER: _____

161. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:

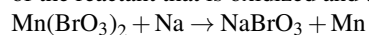


ANSWER: _____

162. Calculate the number of iodine atoms in a 0.22-g sample of strontium iodide (SrI_2).

ANSWER: _____

163. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: _____

164. Suppose Nick heats a sample of a certain chromium (III) chromate hydrate, $\text{Cr}_2(\text{CrO}_4)_3 \cdot 6\text{H}_2\text{O}$ until all the water is removed and pure $\text{Cr}_2(\text{CrO}_4)_3$ is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

165. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

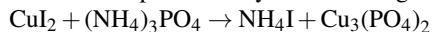
166. Andrew needs to make 232 mL of aqueous potassium sulfide (K_2S) working solution. He plans to do this by adding distilled water to a 53.012-mL stock solution of 0.24-M aqueous potassium sulfide. What will be the concentration of his working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

167. As part of her routine quality-control checks, Victoria, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of nitric acid solution. She takes a 40.-mL sample of nitric acid solution and titrates it with a cobalt (II) hydroxide solution, which she made by dissolving 1.26 g of solid cobalt (II) hydroxide in 1.188 L of distilled water. If Victoria finds that it takes 193 mL of the cobalt (II) hydroxide solution for the titration to reach the equivalence point, what molarity of the nitric acid solution does Victoria calculate?

ANSWER: _____

168. A way to produce solid copper (II) phosphate is by combining aqueous copper (II) iodide with aqueous ammonium phosphate. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Shiv pours 20.03 mL of a 0.06293-M solution of aqueous copper (II) iodide and then adds 14.13 mL of a 0.179-M solution of aqueous ammonium phosphate. He then stirs the mixture and notices copper (II) phosphate precipitating out of the solution. What is the mass of copper (II) phosphate that he should expect the reaction to yield?

ANSWER: _____

169. Of the following labels, identify all that can be applied to the chemical compound Si_3Sb_4 : ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

170. Predict the products that form when carbonic acid is mixed with magnesium hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

171. Nick measures the amount of arsenic trichloride produced during a chemical reaction. He finds that 0.8469 mol is produced. Calculate the mass of arsenic trichloride produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

172. Which element is 46% (by mass) of magnesium sulfite (MgSO_3)? Show your work.

ANSWER: _____

173. A brick has a mass of 8.4×10^0 kg and a virus has a mass of 2.8×10^{-11} mg. What is the mass of 1 mole of bricks? Ensure your answer has the correct number of significant figures.

ANSWER: _____

174. Kevin weighs out 2.901 g of chlorous acid (HClO_2) into a 150.-mL volumetric flask and dilutes the solution to the mark with distilled water. He plans to titrate the acid with 0.3872-M strontium hydroxide solution. Calculate the volume of strontium hydroxide solution that Kevin must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

175. Hannah is instructed to dissolve 1.57 g of lithium fluoride in 82.38 mL of a 82.21-mM aqueous solution of aluminum bromide. Calculate the molarity of the bromide anion in the solution after the reaction has occurred. Assume the volume of the solution does not change when bromide is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

176. A particular planet has a mass of 9.4×10^{26} g and a coin has a mass of 2.5×10^0 g. How many moles of coins have a mass equal to the mass of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

177. What is the oxidation number of tellurium in diarsenic tritelluride?

ANSWER: _____

178. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

179. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

180. Olivia has a 0.08559-M solution of copper (II) iodide (CuI_2). If she needs to add 6.981 g of copper (II) iodide to a reaction flask, what volume of the solution should she pour? Be sure your answer has the correct number of significant figures.

ANSWER: _____

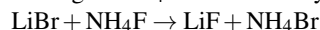
181. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

182. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

183. Given this unbalanced chemical equation, report the maximum mass of NH_4Br that could be produced by the mixing of 8.698 g of LiBr with 6.564 g of NH_4F . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

184. What major chemical species are present in an aqueous solution of 0.66-M silver phosphate?

ANSWER: _____

185. Predict the products that form when hypochlorous acid is mixed with bismuth hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

186. Ammonium chloride reacts with caesium oxide to form caesium chloride and ammonium oxide. Suppose 6.567 g of ammonium chloride is mixed with 31.65 g of caesium oxide. Calculate the maximum mass of caesium chloride that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

187. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

188. Potassium hydroxide reacts with nitric acid to form water and potassium nitrate. Calculate the moles of nitric acid needed to produce 0.0593 moles of water. Ensure your answer has the correct number of significant figures.

ANSWER: _____

189. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 44.9 g.

Compound	Initial Mass	Final Mass
A	21.4 g	9.2 g
B	70.9 g	0 g

ANSWER: _____

190. Of the following labels, identify all that can be applied to the chemical compound SbP: ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

191. Na_3PO_4 is the formula for sodium phosphate. Tyler, a chemical engineer, measures a sample of sodium phosphate and determines that it contains 2.94 moles of phosphorus. How many moles of sodium are in the sample?

ANSWER: _____

192. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

193. A coin has a mass of 8.2×10^0 g and an oil tanker has a mass of 7.14×10^8 kg. How many moles of oil tankers have a mass equal to the mass of a coin? Ensure your answer has the correct number of significant figures.

ANSWER: _____

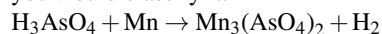
194. Sam has a 0.1017-M solution of hydronium oxide ($(\text{H}_3\text{O})_2\text{O}$). If he needs to add 26.2 g of hydronium oxide to a reaction flask, what volume of the solution should he pour? Be sure your answer has the correct number of significant figures.

ANSWER: _____

195. Matthew must prepare 686 mL of a 0.487-mM working solution of aqueous ammonium sulfide ($(\text{NH}_4)_2\text{S}$). He will do this by pouring some 0.038174-M aqueous ammonium sulfide stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-ammonium sulfide stock solution must Matthew pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

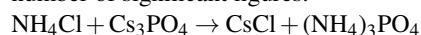
ANSWER: _____

196. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

197. Given this unbalanced chemical equation, report the mass of NH_4Cl needed to produce 10.3 g of CsCl . Ensure your answer has the correct number of significant figures.



ANSWER: _____

198. Ammonium bromide reacts with iron (III) phosphate to form iron (III) bromide and ammonium phosphate. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

199. Chloe prepares a solution of ammonium bicarbonate (NH_4HCO_3) by measuring out 0.5267 g of ammonium bicarbonate into a 100.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

200. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____